# Azhar Amir Kimanje

XR & AI Engineer · Curriculum Vitae

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**Simplicity is Sophistication.** I'm an XR and AI engineer with a unique perspective on improving multi-sensory cognitive decision-making in medical training and challenge-based education.

## EDUCATION

**B.A., Computer Science and Engineering** · Seattle Pacific University, Seattle, WA — Sep 2020 – Jun 2024

## EXPERIENCE

**Lead Interoperability Engineer** · Exonicus Inc., Seattle, WA — Jun 2024 – Present

Real-time XR trauma simulation that trains combat medics to save lives. Also Software Engineer, XR.

- Integrated physiology engines, virtual patients and military training standards (HLA, JETS, MOHSES) so separate simulation systems share one patient in real time.
- Engineered a custom API plugin for real-time medical data communication, getting training to military medics **11 months** sooner.
- Shipped MilExTS (Meta Quest Store) and Exonicus Scenario Builder (Microsoft Store) to put trauma training in medics' and educators' hands.
- Optimized data handling with Unity DOTS to process **thousands of medical injuries** with minimal cache misses, and standardized naming across large medical datasets so teams could use them.
- Led IEEE quality assurance for mixed reality game-engine plugins and designed CI/CD pipelines, integrating the JETS and MOHSES APIs with a team of 5.
- Drove collaboration in a **five-company** Simulation Interoperability Working Group cohort, supporting LVC training commercialized in **six countries**.
- Partnered with US Army DEVCOM on XR usability studies, and taught educators to build no-code AR trauma scenarios at IMSH 2026.

**Director of Mixed Reality and AI Outreach** · AVELA Non-profit, Seattle, WA — Jun 2024 – Present

Director of AI × XR Labs, bringing computing and XR education to underrepresented and rural students.

- Led on-device AI research with Apple MLX so student data stays on the device, shipping the result as CourseSLM on the App Store.
- Secured and executed a **$200K Apple Community Education Initiative award** (2026), an **$80K Apple grant** (2025) and a **$75K UW Student Technology Fee grant** (2024) to scale XR and AI education.
- Orchestrated XR programs in data activism and environmental awareness that reach **3,200+ STEM students a year**, with partners including Apple Education and SPEEA ACE.
- Designed the acquisition pipeline for 30 Quest 3 headsets, 60 laptops and 30 iPads so every student had hands-on access.
- Published a teaching kit on agentic AI coding on Apple's Education Community to help developers and educators adopt AI responsibly.

**Founder & CEO** · Sablesync, Seattle, WA — Nov 2023 – Present

Spatial computing that anyone can use; ML fine-tuning and digital-twin R&D.

- Launched **four Apple apps** that make 3D capture accessible without specialist gear: Sablesync RC, Omni, 3D Reconstruction and Q-AR (see Shipped Apps).
- Prototyped an emergency-evacuation digital twin with Unity NavMesh AI, automating floor plan to 3D model with NVIDIA Omniverse, from R&D to MVP in **one month** with **3 engineers**.
- Led **3 engineers** on real-time AR hardware communication, cutting system integration time **50%** and development cycles **40%**.
- Automated testing for AR hardware components and streamlined deployment with Oculus Integration, reducing firmware errors.

**Technical Mixed Reality Lead Instructor** · SPEEA Aerospace Career Enhancement (ACE), Tukwila, WA — Dec 2024 – Present

- Teach XR with computer-vision models for Gaussian splats, preparing aerospace professionals for spatial computing careers.
- Designed drone-based solutions for spatial planning and agriculture, and lead VR and AR initiatives for aerospace and sustainability projects.

**Software Engineer, XR – Design Intelligence** · IA Interior Architects, Seattle, WA — Jun 2024 – Sep 2024

- Created **10+ XR space plans** in Unity and C#, and verified all built-environment XR models in Unity and Revit, so architects could test spaces before building.
- Researched generative AI solutions in Python to speed up cost estimation, cutting project time **20%**.

**Augmented Reality Developer, Extern** · Snap Inc., Seattle, WA — Mar 2024 – Apr 2024

- Designed Lens Studio 5.5 AR experiences for sports businesses, and wrote product requests and system design documents to align customers and stakeholders.

**Lead Feature Engineer, Extended Reality Course Designer** · AVELA Non-profit, Seattle, WA — Nov 2023 – Jun 2024

- Built AVELA's first **40-hour** XR curriculum to make computing approachable, raising student STEM engagement **25%**.
- Trained conversational AI agents in XR as teaching assistants, and delivered **80 hours** of ML, AI and data-activism curriculum.

**STEM Lead Instructor** · AVELA Non-profit, Seattle, WA — Sep 2023 – Nov 2023

- Wrote the grants that brought XR curriculum and hardware into AVELA, and taught Python, data activism and computing for environmental awareness.

**Product Owner, Software Engineering Capstone** · Seattle Pacific University — Sep 2023 – Jun 2024

- Led SPU GradPro, a 3D simulation of the CS degree on WebGL, adding the university's first conversational AI scene with generative-AI NPCs of CS professors; won **1st** Place, Top Poster Presentation.

**Product Management Intern** · YouthForce, Boys & Girls Clubs of America — Nov 2023 – Jan 2024

- Ran **5 beta tests** of the Youthforce XR Lounge with **30 testers** at a time, turning feedback into features for the final release on Meta App Labs.

**Software Engineering Intern** · YouthForce, Boys & Girls Clubs of America — Jul 2023 – Dec 2023

- Led full-stack development of the first XR lounge for the National Boys & Girls Club after winning the Youthforce Coding Challenge, shipping the MVP **60%** faster and doubling rendering to **120 fps**.

**Freshman Counseling Team Lead, then Intern** · SPU Undergraduate Admissions — Dec 2020 – Jun 2024

- Led **12 interns** making **1,200+ outbound calls a week** to help students choose SPU, contributing to a **35%** enrollment increase and a **25%** rise in minority admissions to the School of Engineering.

**Early Insight Diversity Series** · Citi — Mar 2024 – Apr 2024

## RESEARCH & DEVELOPMENT

**TCCC SBIR Phase II and IIE** · Exonicus Inc. · Lead Interoperability Researcher — Jun 2024 – May 2026

Principal Investigator: Dr. Kyle Couperus

- Led research on a real-time XR medical simulation system, improving data interoperability, latency and reliability so military training outcomes reflect the field.

**CourseLLM** · AVELA Non-profit · Principal Investigator — Jun 2025 – Sep 2025

- Led **3 PhD candidates and 6 interns** fine-tuning hardware-constrained language models with Apple MLX for data sovereignty, delivering a data-activism lesson plan with Apple's Community Education Initiative and shipping CourseSLM.

**Gaussian Splats for Storytelling** · AVELA Non-profit · Principal Investigator — Jan 2025 – Aug 2025

- Architected a deep-learning workflow that turns 360° drone video into true-to-depth XR environments, published openly with Bellevue XR Labs under the Washington Open Common Education License.

**Avatar Controller Toolkit (XR ACT)** · ISIA Lab, Université de Mons · Research Contributor — Jul 2024 – Present

Principal Investigator: Dr. Kevin El Haddad

- Advancing real-time facial landmark detection so XR avatars express emotion more accurately and efficiently.

**BioDepot Workflow Builder** · Morphic Lab, University of Washington Tacoma · Research Contributor — Sep 2024 – Present

Principal Investigator: Dr. Ka Yee Yeung

- Contributing to a GUI that auto-installs workflow components so non-programmers can reproduce gene research, with Jupyter and Cytoscape support.

**Wordplay** · LCI Lab, University of Washington · Research Contributor — Sep 2023 – Present

Principal Investigator: Dr. Amy J. Ko

- Contributing to an accessible, multilingual programming platform that makes computing education more equitable.

## PUBLICATIONS

- **Welcome to the Agentic Era: Strategic AI Implementation for Developers.** Author. Teaching kit, Apple Education Community, Community Education Initiative, Oct 2026. CC0 license.
- **From Toy to Tool: An Investigation on Student AI Use Cases and Effects.** Co-author. Presented at the World Engineering Education Forum & Global Engineering Deans Council (WEEF & GEDC) 2026, Cartagena, Colombia.
- **Tool or Toy? Student Views on ChatGPT in Culturally Responsive Computing Education: A Preliminary Study.** IEEE Xplore, 2025.
- **Gaussian Splats for Storytelling.** OER Commons, with Bellevue College XR Lab, 2024.
- **Lessons Learned from Creating a Standard for Medical Simulation Interoperability.** Exonicus, 2024.

## SHIPPED APPS

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|---|---|
| **CourseSLM** · On-device AI for STEM learning · AVELA | Apple App Store |
| **MilExTS** · XR trauma-care training for military teams · Exonicus | Meta Quest Store |
| **Exonicus Scenario Builder** · Trauma scenario authoring · Exonicus | Microsoft Store |
| **Sablesync RC** · LiDAR room capture to blueprints and 3D models · iPhone, iPad, Mac, Vision Pro | Apple App Store |
| **Sablesync Omni** · Photos to USDZ 3D models · iPhone, iPad, Vision Pro | Apple App Store |
| **Sablesync 3D Reconstruction** · On-device 3D reconstruction from images · Mac | Mac App Store |
| **Sablesync Q-AR** · USDZ model and animation viewer · iPhone, iPad, Mac | Apple App Store |

## GRANTS LED AND EXECUTED

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|---|---|
| **Exonicus Inc.** · XR trauma simulation and interoperability R&D · May 2024 | **$1,900,000** |
| **Exonicus Inc.** · XR trauma simulation and interoperability R&D · Sep 2025 | **$750,000** |
| **Apple Community Education Initiative** · On-device AI and AI education outreach across Washington State · Summer 2026 | **$200,000** |
| **Apple** · On-device AI development for the AVELA community · Summer 2025 | **$80,000** |
| **UW Student Technology Fee (STF)** · Expanding XR education across UW and Seattle · Sep 2024 | **$75,000** |
| **UW Student Technology Fee (STF)** · XR instruction and hardware · Feb 2024 | **$65,000** |
| **Total** | **$3,070,000** |

## AWARDS

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| **1st Place, University of Washington Tech Showcase** · Jun 2024 | **$1,000** |
| **1st Place, Top Poster Presentation Award** · Seattle Pacific University · Jun 2024 | **Winner** |
| **3rd of 200, Global Eduhacks** · University of Washington · Prebrief · Jun 2024 | **$500** |
| **1st, Snap Inc. Master of Augmented Reality** · Apr 2024 | **Internship interview** |
| **6th of 150, Amazon AWS & Inrix Hackathon** · Wanderus · Dec 2023 | **Internship interview** |
| **1st Place, Youthforce Coding Challenge** · National Boys & Girls Club · Jul 2023 | **$700** |

## SCHOLARSHIPS

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| **Director's Grant** · Seattle Pacific University · 2020 – 2024 | **$8,000 / year** |
| **Washington Bridge Grant** · Seattle Pacific University · 2020 – 2024 | **$3,000 / year** |
| **Culture Tech Legacy Google, Masterclass and LinkedIn Learning Scholarships** · 2024 | **Fully paid certifications** |

## MEDIA COVERAGE

- **AVELA researchers present at WEEF & GEDC 2026**, AVELA UW, Oct 2026. Co-author of the presented paper.

- **Welcome to the Agentic Era**, Apple Education Community, Oct 2026. Author of the teaching kit.
- **Startup Spotlight Sessions, Seattle Startup Hall**, Exonicus, Aug 2026. Represented Exonicus as Lead Engineer.
- **Exonicus at the IMSH 2026 Vendor Showcase**, HealthySimulation.com video, Jan 2026. On-camera product walkthrough.
- **Workshop: From Zero to Scenario**, IMSH 2026, San Antonio, Jan 12, 2026. Instructor.
- **Exonicus exhibits at IMSH 2026, Booth #104**, Jan 2026. On-site team.
- **Gaussian Splat for Storytelling**, XR Lab @ Bellevue College, YouTube, Sep 2025. Featured describing my drone-based Gaussian splat approach.
- **FIRST Responder SBIR Phase II Touchpoint**, Fort Gordon, Georgia, Jul 2025. Demonstrated the Exonicus Trauma Simulator.
- **Meet the Class of 2024**, Seattle Pacific University Commencement, Jun 2024. Profile.
- **Live@5 Guest List**, SynED. Guest.

## LEADERSHIP

- **Regional Executive Board Finance Chair**, National Society of Black Engineers, Puget Sound region, Jul 2024 – Nov 2024. Managed corporate partnerships and conference financial planning ($160,000).
- **Founder & President**, NSBE Seattle Pacific University Chapter, Sep 2023 – Jun 2024.
- **Program budget**, Director of Mixed Reality Outreach, AVELA Non-profit, Jun 2024 – Present ($150,000).

## ORGANIZATIONS

- Simulation Interoperability Standards Organization (SISO), Medical Modeling & Simulation Community, through Exonicus, 2025 – Present.
- Alliance for OpenUSD (AOUSD), through Sablesync, 2025 – Present.
- Apple Community Education Initiative (CEI), AI education partner, through AVELA.
- Northwest XR EDU Network (NW XR EDU), through AVELA, 2024 – Present.

## CERTIFICATES

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| **Microsoft** | Azure Digital Twins; Connect IoT Data to Mixed Reality with Azure Digital Twins and Unity; Introduction to Azure Digital Twins; Mixed Reality and Azure Digital Twins; Office Specialist: Excel Associate (2024) |
| **Unity** | VR Development; Shaders and Materials; Data-Oriented Design and Entity Component Modeling; Junior Programmer (2023 – 2024) |
| **LinkedIn Learning** | CISSP: Identity and Access Management; Generative AI for Product Managers; Web Programming Foundations; C# Development (2023 – 2024) |
| **Google** | Connect and Protect: Networks and Network Security; Foundations of Cybersecurity; Play It Safe: Manage Security Risks (2024) |
| **Meta · Snap · Exonicus · Other** | Meta Spark Foundations of AR (2024); Snap Master of AR Development (2024); OPSEC Awareness for Military Members and DoD Employees (2024); 3D Graphics on the Web with Three.js |

## SKILLS

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| **Programming** | C# (5+ years), C++, Swift, Python, C, SQL, Java |
| **Spatial & 3D** | Unity (DOTS, NavMesh), Unreal, OpenXR, RealityKit, ARKit, ARCore, USDZ / OpenUSD, LiDAR capture, Gaussian splatting, NVIDIA Omniverse, three.js, .NET |
| **AI & vision** | Apple MLX fine-tuning, small language models, computer vision, deep learning, NLP, generative AI, rapid prototyping |
| **Simulation** | HLA, JETS, MOHSES, real-time data synchronization, CI/CD |
| **Spoken languages** | Luganda (native), English (native), Spanish (elementary) |

## VOLUNTEER WORK

- XR Technical Support and Student Volunteer, National Boys & Girls Club (current) · Scholarship Essay Support and Coding Instructor, AVELA Non-profit (current) · Welcome Team, Seattle Pacific University (2023 – 2024).